Bijective Proofs

1. Use bijective method to prove Fermat's Little Theorem and Wilson's Theorem.

2. Use bijective method to prove Lucas's Theorem.and Kummer's Theorem.

3. Given a positive integer n, prove that the amount of positive integers with digital sum $2n$ and every digit chosen from $\{1,3,4\}$ is a square.

4. The Perrin sequence 0, 2, 3, 2, 5, 5, 7, 10, ... satisfies $a_{n+3} = a_{n+1} + a_n$. Prove that if n is prime, then n divides a_n.

5. Does there exist positive integers $k < m < n$ such that $\binom{n}{m}$ and $\binom{n}{k}$ are coprime?

6. Given a set $M = \{1,2,\cdots,n\}$. An ordered pair of subsets of M (S,T) is called *good*, if every element of S is larger than the size of T and every element of T is larger than the size of S (one or both of S and T can be empty). Prove that the amount of good pairs is the Fibonacci number F_{2n+2}.

7. Prove that $\sum_{k=0}^{58} \binom{2017+k}{58-k}\binom{2075-k}{k} = \sum_{p=0}^{29} \binom{4091-2p}{58-2p}$.

8. Catalan number $c_n = \frac{1}{n+1}\binom{2n}{n}$, evaluate $\sum_{n=0}^{\infty} \frac{c_n}{4^n}$.

9. A sequence $\{a_n\}$ satifies $a_0 = 1$, $a_{2i+1} = a_i$, $a_{2i} = a_i + a_{i-2^e}$ where $e = v_2(i)$.

Prove that $\sum_{i=0}^{2^n-1} a_i = c_{n+1}$ where c denotes Catalan numbers.

10. Given a positive integer n, find the amount of permutations of 1 through n such that there are no three indices $i<j<k$ such that $a_i<a_j<a_k$.

11. On cards you write integers $a_1, a_2, \ldots, a_n$ that are ≤ 1 with positive sum (one example is 1, 1, 1, 0, 0, 0, -2.) Cards are considered distinct even if they have the same number. Call a set of cards *interesting*, if they sum to 1. For each interesting set of cards, write the number $(k-1)!(n-k)!$ on the blackboard, where k is the amount of cards in this set. Prove that the sum of numbers on the blackboard is $n!$. (In this example, you write 0!6! 3 times, 1!5! 9 times, 2!4! 9 times, 3!3! 4 times, 4!2! 3 times, 5!1! 3 times, 6!0! 1 time, and $720\times3+120\times9+48\times9+36\times4+48\times3+120\times3+720=5040$.)

12. You have n cards, each writing a different real number. The sum of all numbers is irrational. A permutation of these cards is good if when reading from left to right, all partial sums are irrational. Find the minimum amount of good permutations.

13. Given a positive integer n, write out all sequences of positive integers with n entries that begins with 1 and each term from the second term on does not exceed "the sum of its predecessor and 1". For example, when n=3, we have 5 such sequences: 111, 112, 121, 122, 123. Then calculate the product of each sequence and add them together, in this case it is 1×1×1+1×1×2+1×2×1+1×2×2+1×2×3=1+2+2+4+6=15. Prove that the result is the product of the first n odd numbers.

14. Let m and n be positive integers, m>n. Let S be the set of ordered n-tuples of positive integers $(a_1, a_2, \cdots, a_n)$ with sum m. Prove that

$$\sum_{(a_1,\cdots,a_n \in S)} 1^{a_1} 2^{a_2} \cdots n^{a_n} = \sum_{i=1}^{n} (-1)^{n-i} \binom{n}{k} i^m \text{ 。}$$

15. Let $\{b_n\}$ be a given sequence of integers, we define $\{a_n\}$ as: $a_1 = b_1$, and $a_n = nb_n + a_1 b_{n-1} + \cdots + a_{n-1} b_1$ for $n \geq 2$. Prove that $a_p \equiv a_1 \pmod{p}$ for prime p.

16. Given a prime number $p \geq 5$. $a_1, a_2, \cdots, a_p$, a permutation of 1, 2,..., p, is called "good", if we have $a_k \neq k$ for k=1, 2, ..., p, and $a_1 + 2a_2 + \cdots + pa_p$ is divisible by p.

Find the remainder of the amount of good permutations divided by $p(p-1)$.

17. Given a positive integer n and a prime number p, prove that $n \left| \binom{n+p-1}{p} - \binom{n}{p} \right.$.

18. Evaluate $\sum_{k=0}^{n} (-1)^k \binom{2n-k}{k}$.

19. Prove that $n!(n+1)!(n+2)! \mid (3n)!$ for integer $n \geq 3$.

20. Prove that $\binom{n}{p}\binom{n}{q} = \sum_{k=0}^{n} \binom{p}{k}\binom{q}{k}\binom{n+k}{p+q}$.

21. Prove that $\sum_{k=0}^{\left\lfloor \frac{n-1}{2} \right\rfloor} (-1)^k \binom{n+1}{k}\binom{2n-1-2k}{n} = \frac{n(n+1)}{2}$.

22. Prove that $\sum_{k=0}^{\left\lfloor \frac{n}{2} \right\rfloor} (-1)^k \binom{n}{k}\binom{2n-2k}{n} = 2^n$.

23. Prove that $\sum a_1 a_2 \cdots a_k = F_{2n}$, the left side sums over all integer compositions of n.

24. Prove that $\sum (2^{a_1-1}-1)(2^{a_2-1}-1)\cdots(2^{a_n-1}-1) = F_{2n-2}$, the left side sums over all integer compositions of n.

25. Prove that $\sum_{k=0}^{n} \binom{2k}{k}\binom{2n-2k}{n-k} = 4^n$.